

4. The normal to the circle $x^2 + y^2 = 4 - 2x$ at the point $A(1, -1)$ passes through another point B on the circle.

(a) Find the centre and radius of the circle. [3]

(b) Find the equation of the tangent to the circle at point B . [6]

5. (a) Solve the equation $\log_2(3x-1) = 1 + \log_4\left(x^2 + \frac{9}{4}\right)$. [4]

(b) Solve the equation $2e^x - 5 = 3e^{-x}$.

[4]